SwissSigma: Scientist instructions

Numbered items are manual tasks.

Thank you for your work and helping us participate in this hack!

▼ [A] File structure and set up

0. Please clone our repo (https://github.com/zahmiriajoh/SwissSigma-SDLhack)

```
#these are the folders + files relevant to the physical ru
n
SwissSigma-SDLhack/
├── setup.sh
│
└──  production/
    ├── run_bo.sh
    ├── sdl_solubility_ot2.py
    ├── deck_map.jpg
    ├── plate_layout.jpg
    └── csvs/
        ├── ratios/
        └── uvvis/
```

You will need to upload to the Opentrons:

- any novel labware definitions (see OT protocols/labware/)
- sdl_solubility_ot2.py
- the round's associated ratio instructions
    - When you set up a run in the App, the parameters screen shows a **Round conditions (CSV)** field. You click it and pick that round's

`ot_input_roundN.csv` from the repo on the computer.

- The App sends the protocol and that CSV to the robot together. The protocol runs on the robot's own computer, not the laptop, so it can only see files the App sent.
- On the robot, `protocol.params.input_csv.parse_as_csv()` reads the uploaded copy.

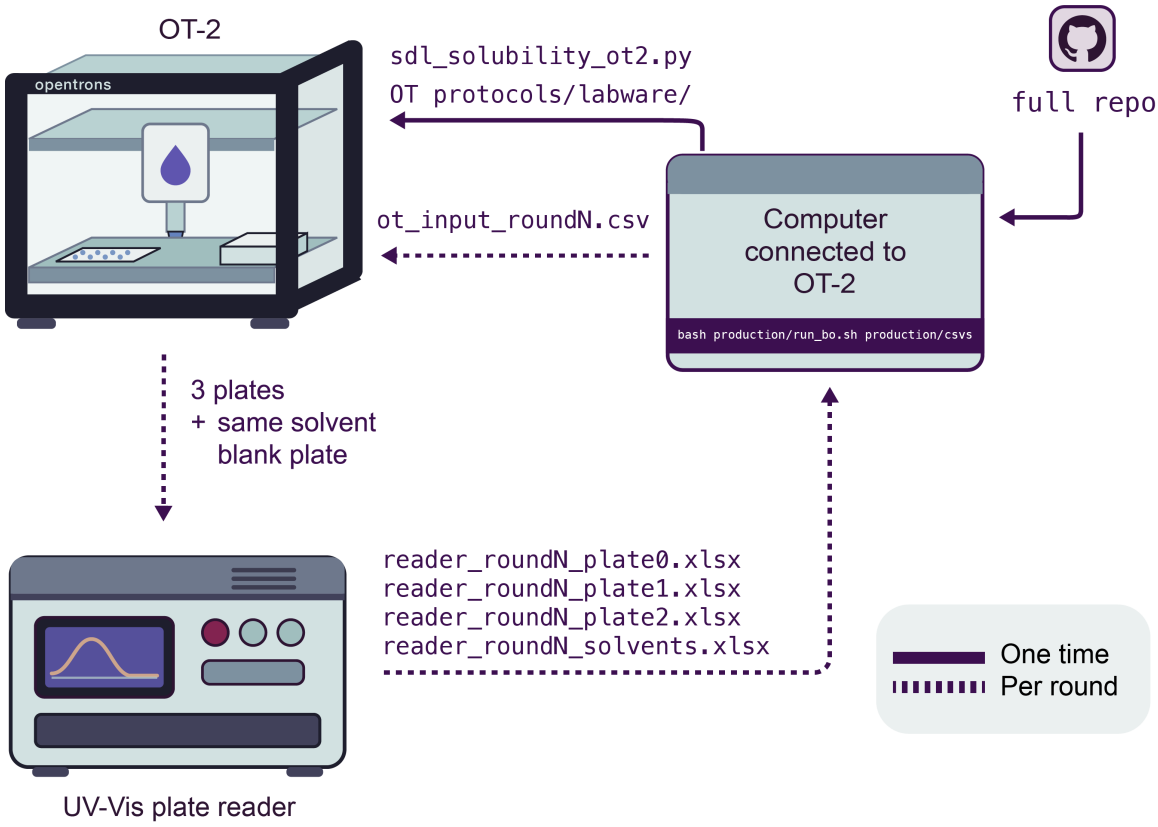
The BO analysis will require the installation and activation of a specific environment

```
#installing uv and building the venv (do once per compute
r)
bash setup.sh

#This installs uv if needed, builds the environment in .ve
nv/, and runs a quick self-check. It ends with "All good."
```

```
# When running scripts, if uv isn't found, open a new term
inal and run
# bash setup.sh again.
```

On the very first run, with an empty folder, the BO command in step 9 writes the round 0 starting design instead. If the exports are missing, it lists the files it is waiting for. It is safe to re-run, because it never overwrites a finished round.



0. We are wanting to quickly pre test what the low grain solubility is for each solute in DMF as a baseline

For each of the four solutes, please test 5, 10, 20 and 50 mg/mL solute in DMF

5 minutes of shaking is sufficient, if needed.

Please use the highest concentration (in 300 uL of solvent) as the amount of solids to add per well in step 1 below.

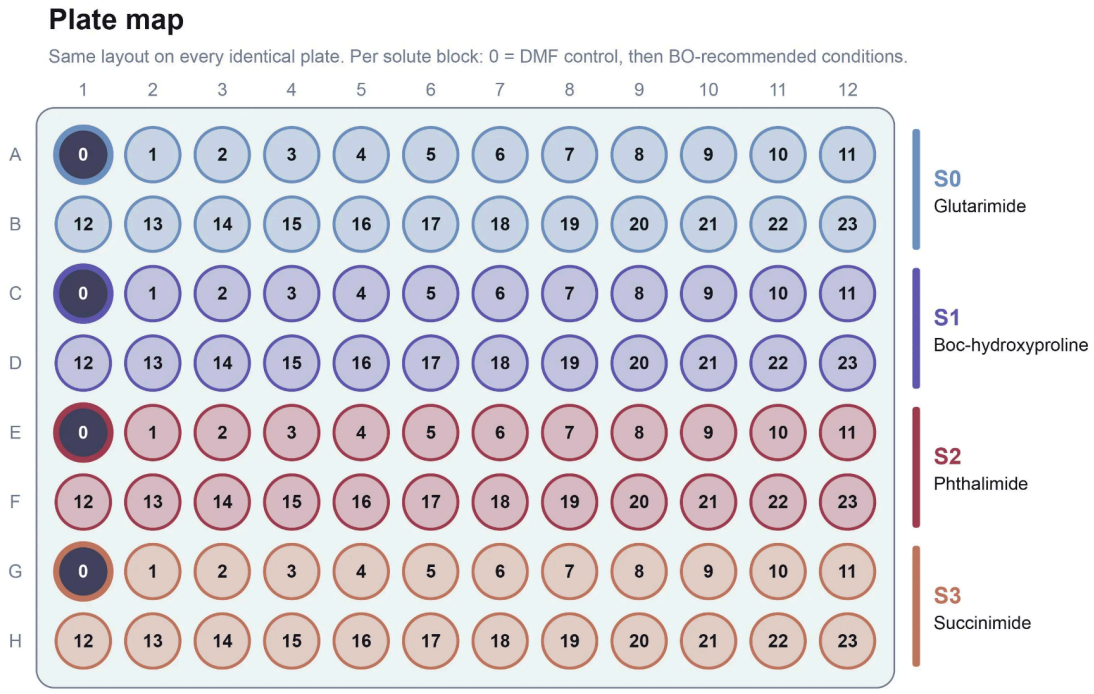
▼ [B] deck layout and starting configuration

To evaluate solubility, a researcher will prepare the solids in the well plate and place it into the Opentrons system.

Manual prep needed from scientists:

1. Weigh out solids into the plates according to the plate map below 9 times (3 rounds of 3 plates each)

Solute	Amount to add per well (mg)
Glutarimide	see step 0 above
Boc-hydroxyproline	
Phthalimide	
Succinimide	



2. Place modules and plates according to the deck map below

OT-2 deck map: to run sdl_solubility_ot2.py

P300 GEN2 single (left) · P20 GEN2 single (right)



Solvents:	**Amount to fill per trough per round (mL) (2 troughs, 3 rounds)**
DMF	30
Acetone	30
EtOH	30
i-PrOH	30
Water	30
i-PrOAc	30
t-BuOAc	30

The robot will fill the solvent blank plate.

Naming convention: round0_plate0_solute0_cond0

- 3 rounds (0-2)
- 3 plates per round (0-2)
- 4 solutes (0-3)
- conditions (0-23)
  - 1 solute in DMF control

Note that plate 0 and solute 0 exist! So A1 = roundN_plateN_solute0_cond0

The 6 solvents + DMF will be stored in a 8-trough plate.

▼ [C] moving and mixing liquids

The Opentrons system will then dispense the varying green solvent mixtures into a well plate. After a specified agitation/incubation period, the UV-Vis absorbance will be measured.

The total solvent added to a well will be 300uL, consisting of 1-3 solvents, adding varying ratios of each.

- **Filling:** all three are filled in one pass, with each aspiration serving the same well on plates 0, 1 and 2 back to back.
- **Tips:** a full round uses only about 7 tips, because each tip only touches its own solvent.

The ratios instruction CSV will be stored at `production/csvs/ratios/` .

3. Press start on the run.

4. Once the liquid adding is done, the run will pause. Please add seals to each of the 3 solute plates.
5. Move the plates one at a time to the shaker and resume the run for each.

- **Shaking:** each plate is shaken in turn. The app prompts you to move it onto the heater-shaker in slot 10 and back, since the OT-2 has no gripper.
- The 3 plates will one at a time be moved to the shaker for 20 min and no heating (RT is part of the green solvent use in PROTAC synthesis motivation), then taken out for readout on the plate reader.

6. When the last plate finishes shaking, the run should be done. Please read the plate by UV-Vis.

▼ [D] UV measurement

The primary analytical equipment will be a UV-Vis plate reader; high absorbance indicates high kinetic solubility of the active compounds, while low absorbance indicates precipitation or failure to dissolve.

7. The plates will be moved to a plate reader for readout of the full spectrum (200-700nm), with a step size of 5nm.

Include measurement of the solvent blanks plate in grid 6 and return the solvent blanks plate to the OT-2.

8. Please upload the UV-Vis data to `production/csvs/uvvis/`

Name the 4 .xlsx files with the following convention:

File	Data on
`reader_roundN_plate0.xlsx`	sample plate 0
`reader_roundN_plate1.xlsx`	sample plate 1
`reader_roundN_plate2.xlsx`	sample plate 2
`reader_roundN_solvents.xlsx`	the solvent blank plate (slot 6), requiring the N+1 column is filled

The four plate files are required. If any is missing, `run_bo.sh` stops and lists them.

▼ [E] data output and feed into BO

9. With the UV-Vis XLSXs uploaded, run this command from the repo root:

```
bash production/run_bo.sh production/csvs
```

It analyzes the spectra, runs the BO, and writes the next round's instructions into `production/csvs/ratios/`

Non-compatible solvents are

- i-PrOAc with water
- t-BuOAc with water

These combinations will not be tolerated in the BO

▼ [F] input from BO and successive rounds

10. Between rounds, the 3 plates of solids will have to be replaced and the solvent reservoir(s) will need to be refilled.
11. Re-run the protocol on the OT.

▼ [G] ending configuration and final collected data

The deck should look the same as it started, with the reservoir depleted.

And we should have the following 6 sets of data for analysis in csvs/:

- 3 sets of UV-Vis results
- 3 sets of BO-recommended conditions (see step 9)
    - 1 generated with the BO commands for 1st round conditions

- 2 generated from responding to collected data